



Optimization Strategies for Airplane Boarding

LORENTZIAN GEOMETRY AND
ITS APPLICATIONS

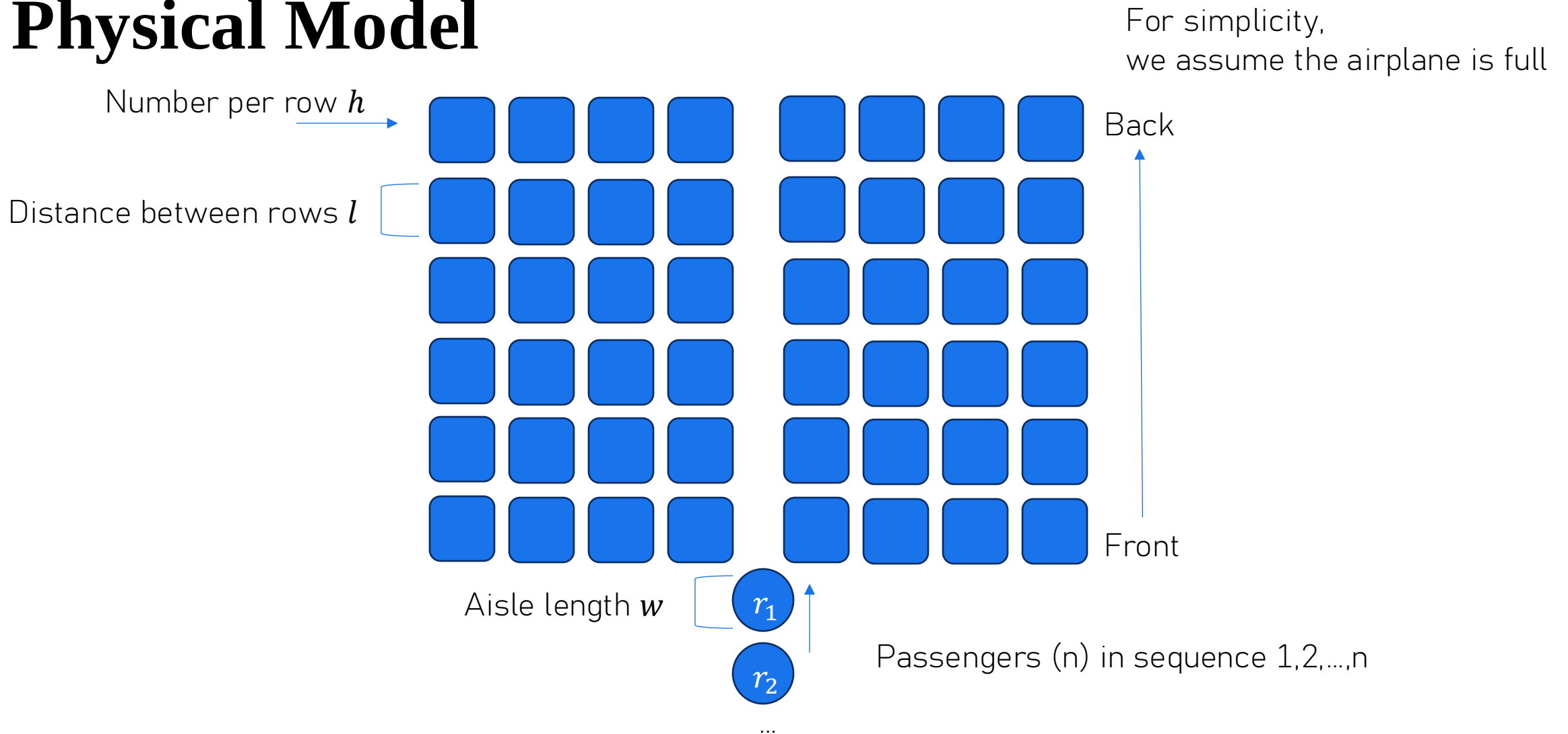
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Introduction

- Motivation & Real-World Problem
- Geometric Connection (Lorentzian Geometry)

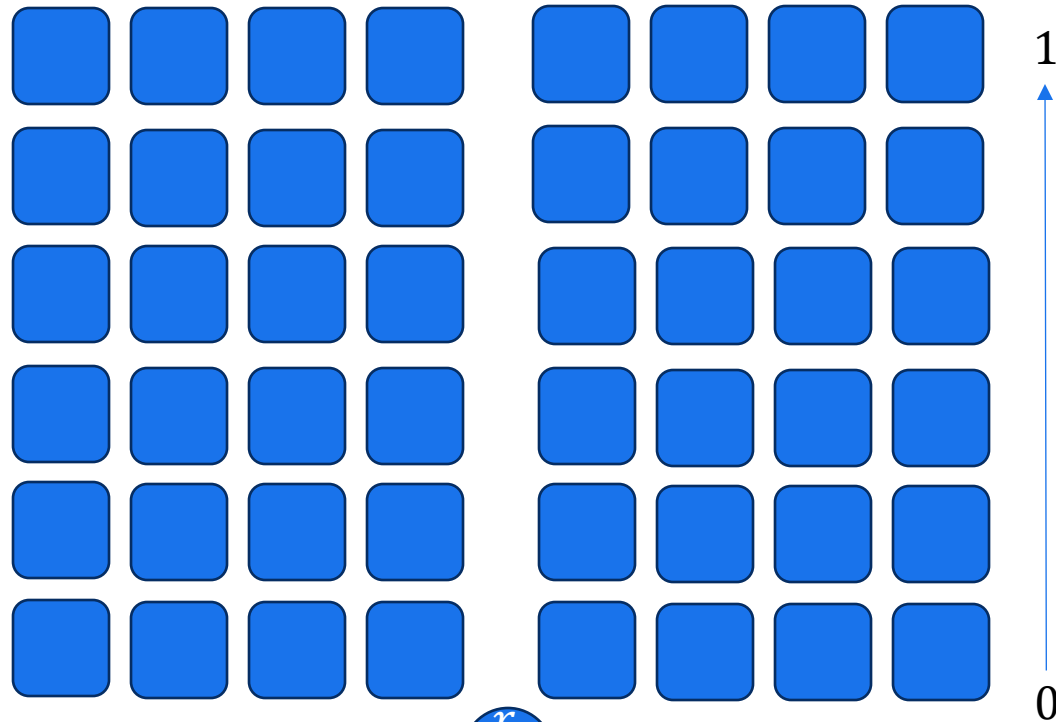
Physical Model



Physical Model

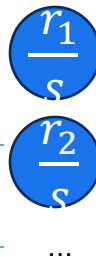
Congestion parameter k

$$k = \frac{hw}{l}$$



Blocking potential

$$\alpha(q, r) = \int_0^1 p(q, z) dz$$



Normalize queue location and row location $(\frac{i}{n}, \frac{r_i}{s})$

Mathematical Framework

Quadratic form

$$ds^2 = \underbrace{4p(q,r)}_{\text{Normalizing factor}} (k\alpha(q,r)dq^2 + dqdr)$$

Normalizing factor

Boarding metric

$$M(q,r) = 4p(q,r) \begin{bmatrix} k\alpha(q,r) & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$$

Expected people between a queue difference dq:

$$\begin{aligned} E &= n \int_{dq} \int_r^1 p(q,r) dr dq \\ &= n\alpha(q,r) dq \end{aligned}$$

And X is blocking X' if

$$dr \geq -\frac{kE}{n} = k\alpha(q,r) dq$$

Therefore, we are able to get the metric.

Mathematical Framework

$$L(\phi) = \int ds = \int_0^1 \sqrt{4p(q, \phi(q))(k\alpha(q, \phi(q) + \phi'(q)))} dq$$

Euler-Lagrange equations

$$\frac{\partial L}{\partial x^\mu} - \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^\mu} \right) = 0$$

Beltrami equation

$$H(x) = \frac{y' \partial F}{\partial y'} - F = c \text{ where } F \text{ is the functional}$$

Connection with Continuous Space

Theorem. Let D be a nice domain of unit volume. Let $\text{diam}(D)$ denote the maximal length of a causal curve in D . Let $L_n(D)$ be the maximal length of a chain among n points, sampled according to the volume distribution on D . Let $X_n(D)$ be the total number of chains of all sizes among n sampled points. Then,

1. For all $\varepsilon > 0$, with high possibility, we have

$$\left| L_n(D) - c_d \text{diam}(D) n^{\frac{1}{d}} \right| < \varepsilon n^{\frac{1}{d}}$$

Moreover, with high possibility, given a maximal chain C , there is a maximal curve γ_C such that all points $x \in C$ satisfy $d_S(x, \gamma_C) < \varepsilon$, i.e. the maximal chain will be contained in an ε neighborhood of a maximal curve.

Connection with Continuous Space

Theorem. (continued)

2. For all $\varepsilon > 0$, with high possibility, we have

$$\left| \log(X_n(D)) - \rho_d \text{diam}(D) n^{\frac{1}{d}} \right| < \varepsilon n^{1/d}$$

where ρ_d depends only on the dimension d .

Moreover, given a uniformly random chain C , with high possibility, there is a maximal curve γ_C such that all points $x \in C$ satisfy $d_S(x, \gamma_C) < \varepsilon$, i.e. with high possibility, a randomly chosen chain is contained in an ε neighborhood of a maximal curve.

Connection with Continuous Space

Theorem. Let $L_{n,h,l,w,p}$ be the boarding time random variable and $L_{n,k,p}$ be the longest chain among n points sampled from $S_{k,p}$. Then, for all $\varepsilon > 0$, with high possibility we have

$$|L_{n,h,l,w,p} - L_{n,k,p}| \leq \varepsilon\sqrt{n}$$

and hence

$$|L_{n,h,l,w,p} - D(S_{k,p})\sqrt{n}| \leq \varepsilon\sqrt{n}$$

Where $D(S_{k,p})$ denotes the diameter of the space-time domain $S_{k,p}$.



Mathematical Framework (continued)

Definition. The **normalized boarding time** of a boarding policy F given by p and a congestion parameter k is the diameter $Diam(S_{k,p})$ and will be denoted by $B(F, k)$.

Parametrize causal curve ϕ :

$$r = \phi(q)$$

Analysis of Boarding Strategies

Random Boarding Policy

Policy F_{ran}

Blocking potential $\alpha(q, r) = 1 - r = 1 - \phi$

Normalized boarding time $B(F, k)$

$$L(\phi) = \int_0^1 2\sqrt{\phi'(q) + k(1 - \phi)} dq$$

Beltrami equation $H(x) = \frac{y' \partial F}{\partial y'} - F = c$ where F is the functional

Analysis of Boarding Strategies

General solution for $\phi(q)$

$$\phi(q) = c_1 e^{2kq} + c_2 e^{kq} + 1$$

Normalize with initial condition $\phi(0) = 0, \phi(1) = 1$

$$\phi(q) = \frac{e^{2kq}}{e^k - 1} - \frac{e^k}{e^k - 1} e^{kq} + 1$$

Solution

$$B(F_{ran}, k) = \text{diam}(S_k) = 2 \sqrt{\frac{e^k - 1}{k}}, \quad k \leq \ln(2)$$

Analysis of Boarding Strategies

Prop. Maximum curve consists of a line segment ϕ_1 in $q \in [0, q_0]$, followed by a curve ϕ_2 which solves the Euler-Lagrange equation between point $(q_0, 0)$ and $(1,1)$.

Proof.

Consider $\tilde{L}(\phi) = 2 \int_0^1 \sqrt{\phi' + k} dq \geq L(\phi)$. The Euler-Lagrange equation associated with $\tilde{L}$ is $\phi'' = 0$.

For $\phi(q) = 0$ the values of L and $\tilde{L}$ coincide.



Analysis of Boarding Strategies

Boundary condition

$$\phi_2'(q_0) = 0, \phi(q_0) = 0, \phi(1) = 1$$

$$q_0 = \frac{k - \ln 2}{k}$$

Solution

$$B(F_{ran}, k) = 2 \left(\sqrt{k} + \frac{1 - \ln 2}{\sqrt{k}} \right), \quad k \geq \ln(2)$$

Analysis of Boarding Strategies

Back-to-Front Boarding Policy

Suppose groups are divided in $S_1, \dots, S_m$ with endpoints $r_0, r_1, \dots, r_m$.

For $(q, r) \in S_i$,

$$p(q, r) = \frac{1}{r_{i-1} - r_i}$$

$$\alpha(q, r) = \frac{r - r_i}{r_{i-1} - r_i}$$

$$ds^2 = \frac{4}{r_{i-1} - r_i} \left(k \frac{r - r_i}{r_{i-1} - r_i} dq^2 + dqdr \right)$$

Analysis of Boarding Strategies

For simplicity, consider we split the passengers into m equally sized groups:

$$\begin{aligned} p(q, r) &= \varepsilon, & \text{Above } S_i \\ p(q, r) &= 0, \alpha = 1, & \text{Below } S_i \end{aligned}$$

$$\phi(q) \geq -\left(q - \frac{i-1}{m}\right)k + \frac{m-i}{m}, \quad \text{since bottom left point of } S_i \text{ is given by } \left(\frac{i-1}{m}, \frac{m-i}{m}\right)$$

$$\phi\left(\frac{j}{m}\right) \geq 1 - \frac{j}{m}, \quad \text{for } j = i + \frac{1}{1-k} + 1$$

$$\text{Length of maximal curve in } S_i = \frac{B(F_{ran}, k)}{\sqrt{m}}$$

$$\text{Normalized boarding time} \leq \left(\frac{1}{1-k} + 1\right) \frac{B(F_{ran}, k)}{\sqrt{m}}$$

Analysis of Boarding Strategies

Theorem. If $k > 1$, then for any partition $\bar{r}$, we have $B(\bar{r}, k) \geq 2\sqrt{k-1}$. Consequently, the boarding time of any back-to-front policy is at most $\frac{\sqrt{k}(1+\frac{\ln 2}{k})}{\sqrt{k-1}}$ times faster than that of random boarding.

Analysis of Boarding Strategies

Denote $x = 1 - r_1$ and v number of rows in the airplane, two boarding groups:

$$S_2 = 1, \dots, [(1 - x)v] \text{ and } S_1 = [(1 - x)v] + 1, \dots, v$$

Proper time of maximal curve that contains entirely within region i : $T_i(x, k)$

$$T(x, k) = \max\{T_1(x, k), T_2(x, k), T_{1,2}(x, k)\}$$

For $k = 1$, local minimum $x \approx 0.5$

For $k > 1$, local minimum: one at closer to $x = 0$ and one closer to $x = 1$



Conclusions & Discussion

Practical Insights & Limitations

Broader Implications: stadium exits, hospital workflows



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